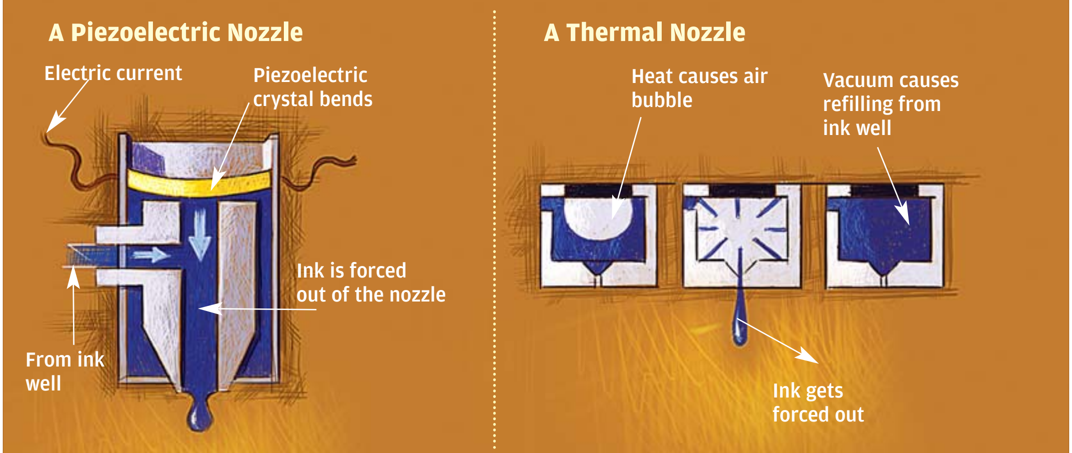
How Inkjet Nozzles Work



There are two types of inkjet nozzles—piezoelectric, on the left, and thermal, on the right. In the former, a current bends the crystal, which forces the ink out. In a thermal nozzle, it's based on the formation of an air bubble.

3D Printing

Going beyond 2D inkjets, think about 3D printers. A 3D printer is essentially a modelling tool, which can create plastic moulds and such. As an example, Z Corp, based in Burlington, Massachusetts, USA, markets an "affordable 3D printing system", which uses a spray nozzle adapted from an HP inkjet printer, to spray a liquid that binds powdered solid substances into a certain shape.

Z Corp began shipment of its Spectrum Z510 3D printing system in March of this year—the first high-definition colour 3D printer ever to hit the market. The Z510 introduced HD3DP (high-definition 3D printing) for rapid prototyping to designers, engineers, and product developers. (Rapid prototyping refers to a class of technologies that can construct physical models from Computer-Aided Design, or CAD, data.)

Using the system interface, product developers could now efficiently print 3D physical models with smaller features and more complex geometries, in colour, and with high-definition detail. The system software also allows for labelling, enhanced texture mapping, and vibrant product colouring. The Z510 initially retailed at $49,900 (Rs 21 lakh) in the United States.

But such printers can, obviously, be used for more than just modelling and visualisation. It's all about design: you feed in the requisite data to the printer, and it constructs the required object according to the blueprint.

And that is something to think about. You could feed in a blueprint for just about anything! Could such inkjet (and other) printing technologies democratise manufacturing the way the invention of the printing press democratised knowledge hundreds of years ago? Instead of inkjet printers that print just words and graphics on paper, we could all have 3D printers sitting on our desks, printing out whatever we told them to.

And what would that mean in terms of who would make the money? There would be a paradigm shift: you would pay for plans rather than for products. Download a plan, print out the product. You are probably already imagining P2P networks that you could download plans off!

What kinds of product would you really be able to print out? What is feasible in reality? As an indicator, instead of looking at fiction of the future, let's look at what is being done right now.

Flexonics

John Canny, professor of engineering at the University of California, Berkeley, and his co-investigator Vivek Subramanian, are using electroactive polymers, gold nanocrystals and inkjet printers to print devices that can move as well as process information. (Electroactive polymers are substances that respond to external electrical stimulation by displaying a significant displacement in shape or size, and nanocrystals are a non-traditional type of semiconductor.) They call this process 'flexonics'.

This revolutionary approach to desktop manufacturing has been enabled by recent advances in 3-D printers and organic electronics.

Printing Out Houses

Better inhalers for better delivery of drugs. Plastic screens. Tissue engineering. Fuel injection. Rocket spare parts. Bones. Organs. Skin. What is not possible with inkjet technologies? Houses? As it turns out, houses can be printed out, too. One researcher has plans to, as he puts it, "be able to completely construct a one-storey, 2000-square foot home on-site, in one day and without using human hands", by 'printing' the house.

Engineer Behrokh Khoshnevis, at the University of Southern California, has been perfecting his 'Contour Crafter' for more than a year now. It 'prints' houses, and is to be tested by the construction industry. It takes instructions directly from an architect's computerised drawings and then squirts layers of concrete one on top of the other to build vertical walls. The precision automaton could revolutionise building sites, for the simple reason that it can work round the clock, in darkness and without breaks.

The key to the technology is a computer-guided nozzle that deposits a line of wet concrete, something like toothpaste being squeezed out. Two trowels, attached to the nozzle, move to shape the formed deposit. The robot repeats the same step many times to give height; hollow walls are built and the machine returns to fill them.

Degussa AG, of Düsseldorf, Germany, will collaborate on Khoshnevis' project to help him find the best kind of building material. Khoshnevis has tested his prototype with cement, but believes that adobe—a mix of mud and straw dried by the sun—might be suitable.

Khoshnevis' prototype robot hangs from a movable overhead gantry (a large supporting structure). The first house will be built later this year. If the technology proves successful, the robot could enable new designs that cannot be implemented using conventional methods, for example, structures involving complex curving walls.